```
# Naive bayes for wine dataset:
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score,confusion_matrix

data=load_wine()

x=data.data
y=data.target

x_train,x_test,y_train,y_test=train_test_split(x,y,train_size=0.8,random_state=42)

model=GaussianNB()
model.fit(x_train,y_train)

y_pred=model.predict(x_test)
print(y_pred)

test=model.predict([x[0]])
print(test)

acc=accuracy_score(y_test,y_pred)
print(acc*100)

print("Confusion Matrix")
print(confusion_matrix(y_test,y_pred))
```

```
# SVM for wine dataset:
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix

data=load_wine()

x=data.data
y=data.target

x_train,x_test,y_train,y_test=train_test_split(x,y,train_size=0.8,random_state=42)

model=SVC(kernel="linear",C=1.0)
model.fit(x_train,y_train)

y_pred=model.predict(x_test)
print(y_pred)

test=model.predict([x[0]])
print(test)

acc=accuracy_score(y_test,y_pred)
print(acc*100)

print("Confusion Matrix")
print(confusion_matrix(y_test,y_pred))
```

Outputs:

```
(.venv) PS D:\Internship\Day42> python task.py
[0 0 2 0 1 0 1 2 1 2 0 2 0 1 0 1 1 1 0 1 0 1 1 2 2 2 1 1 1 0 0 1 2 0 0 0]
[0]
100.0
Confusion Matrix
[[14  0  0]
 [ 0 14  0]
 [ 0  0  8]]
[0 0 2 0 1 0 1 2 1 2 0 2 0 1 0 1 1 1 0 1 0 1 1 2 2 2 1 1 1 0 0 1 2 0 0 0]
[0]
100.0
Confusion Matrix
[[14  0  0]
 [ 0 14  0]
 [ 0  0  8]]
```